Open Source Audit Project Report

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Course: Open Source Software

Chosen Software: Git

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1. Introduction

Most of the software that powers the modern world the servers that run websites, the tools developers use daily, the operating systems inside supercomputers and smartphones was not built by a single company for profit. It was built openly, shared freely, and improved collectively. That is the essence of open source.

This project asks me to stop and think about that.

I chose **Git** - a version control system created by Linus Torvalds in 2005. This report conducts a structured audit of Git: why it exists, who built it, what freedoms its license gives, what philosophical values it represents, and how it has shaped the world.

Alongside this report, I have written five shell scripts that demonstrate practical Linux skills automation, package inspection, log analysis, and a manifesto generator.

2. Part A — Origin and Philosophy

2.1 A1: Origin Story

Before Git, the Linux kernel development team used a proprietary distributed version control system called **BitKeeper**. BitKeeper was free for open-source projects, which was generous at the time. But in 2005, the company behind BitKeeper, BitMover, revoked the free license after a developer reverse-engineered the protocol. Linus Torvalds and his team suddenly had no version control system.

The alternatives at the time were:

- **CVS** – Centralized, slow, terrible with merging.
- **Subversion (SVN)** – Better than CVS but still centralized. Every commit went to a central server. Branching was heavy and slow.
- **No system** – Emailing patches back and forth, which is what Linux had done earlier.

Linus wanted something that was:

- **Distributed** – Every developer has a full copy of the repository history.
- **Fast** – Commits, diffs, logs, branching must be instantaneous.
- **Safe** – Cryptographic hashing (SHA-1) to ensure data integrity.

Freedom	Description	Does Git grant it?
Freedom 0	Run the program for any purpose	Yes
Freedom 1	Study and modify source code	Yes — source code is fully available
Freedom 2	Redistribute copies	Yes
Freedom 3	Distribute modified versions	Yes — but under GPL v2, modifications must also be GPL

• GPL vs MIT

- **GPL** (copyleft) – Derivative works must also be open source under GPL.
- **MIT** (permissive) – You can take the code, modify it, close it, and sell it as proprietary. You only need to keep the original copyright notice.

• Commercial usage

For a library or tool that I want to be widely adopted (even by companies), I would choose MIT. For something like Git — critical infrastructure — I would choose GPL v2 to ensure it never becomes proprietary.

• Free vs freedom

- Free beer = zero cost. Git is free of cost.
- Free speech = liberty to modify and share. Git grants that through GPL.
  Example: GitHub offers Git hosting for free (beer), but the freedom to run your own Git server, modify the client, or fork the project — that is freedom.

2.3 A3: Ethics of Open Source

Should all software be open source?

Argument for yes:
Proprietary software hides bugs, backdoors, and insecure code. The public cannot audit voting machines, medical devices, or critical infrastructure. Stallman argues that source code is knowledge, and withholding knowledge is unethical.

Argument against no:
Some software has competitive trade secrets. A game company's rendering engine might be its advantage. Also, open source does not automatically mean sustainable — developers need to eat. Many open-source maintainers burn out with no pay.

Is it ethical for a company to build profit on free community labor?
Yes — as long as they contribute back. Red Hat, Google, and Meta all use Linux. Google employs kernel maintainers. Meta contributes to open-source AI tools. The unethical version is "open-core" companies that take but never give back.

"Standing on the shoulders of giants" in software:
Every programmer today stands on Git (version control), Linux (operating system), GCC/Clang (compilers), and thousands of open-source libraries. Without them, we would write everything from scratch — like building a skyscraper by first inventing concrete.

Does open source enable or undermine innovation?
It enables it. Innovation is not just "first to invent" but "best to refine." Linux did not invent the operating system — it perfected open collaboration. Git did not invent version control — it made distributed VCS fast. Open source allows many minds to solve a problem faster than any single company.

3. Part B — Linux Footprint

• Installation method

sudo apt update

sudo apt install git -y

• Key directories

Type	Path	Description
Binaries	`/usr/bin/git`	Main executable
Config (system)	`/etc/gitconfig`	System-wide config
Config (user)	`~/.gitconfig`	User-specific config
Config (repo)	`.git/config`	Per-repository config
Man pages	`/usr/share/man/man1/git*`	Documentation
Helper scripts	`/usr/lib/git-core/`	Internal Git tools

• User and group

Git runs as the **logged-in user** who executes the command. Git does not have a system daemon (unless using Git daemon or GitLab). This matters for security: if a user has write

access to a repository, Git respects filesystem permissions. Git itself does not add an extra privilege layer.

• Service management

Git is not typically run as a background service. However, you can run git daemon to serve repositories read-only over the network:

sudo systemctl start git-daemon

• Update model

sudo apt update && sudo apt upgrade git

4. Part C — FOSS Ecosystem

• Dependencies

Git is written in C and shell script. Its core dependencies:

- **libcurl** – For HTTP/HTTPS networking (cloning over HTTP)
- **libssl** (OpenSSL) – For encrypted communication
- **zlib** – For compression
- **pcre/perl** – For regular expressions in git grep
- **expat** – For XML parsing (used by git send-email and some hooks)
- **OpenSSH client** – For SSH-based repository access

• Related tools

- **GitHub** (2008) – The world's largest code hosting platform.
- **GitLab** – Self-hosted alternative.
- **Bitbucket** – Initially Mercurial, now Git-first.
- **Linux kernel development** – Git was built for it and still powers it.
- **Thousands of companies** – Every tech company uses Git.
- **Forks**: libgit2 (pure C implementation), JGit (Java), Go-git.

• Community

- **Mailing list:** git@vger.kernel.org (Linus and Junio Hamano still there)
- **Official documentation:** git-scm.com
- **Conferences:** Git Merge (yearly), Linux Plumbers Conference
- **IRC:** #git on [Libera.chat](https://libera.chat)
- **Governance:** Git is maintained by Junio Hamano (since 2005). No corporate foundation — but heavily used by Google, Microsoft, GitHub, and many others who contribute back.

Role in Ecosystem - For **99% of projects** from a student's assignment to a startup's codebase to a large open-source project like the Linux kernel **Git is the correct choice**. It is fast, free, distributed, and has an enormous ecosystem.

5. Part D — Open Source vs Proprietary

Dimension	Git (Open Source)	Perforce (Proprietary)
Cost	Free	Expensive licensing (free for small teams)
Security (audit)	Anyone can audit code. Security bugs found by community.	Only Perforce engineers see source. Trust is required.
Support and reliability	Community support (Stack Overflow, IRC, mailing lists). No SLA.	Paid support with SLAs. Enterprise contracts.
Freedom to modify	Full — change anything, recompile.	None — you are a user, not an owner.
Community vs corporate control	Maintained by volunteers and corporate contributors (Google, Microsoft). No single company controls it.	Controlled entirely by Perforce Inc.
Performance	Extremely fast for small/medium repos. Slow on monorepos (though Git has partial clone now).	Built for massive monorepos (games, 100GB+).
Binary files	Poor (stores entire binary on every change).	Excellent (only stores diffs).

5.1 Conclusion

For **99% of projects** — from a student's assignment to a startup's codebase to a large open-source project like the Linux kernel — **Git is the correct choice**. It is fast, free, distributed, and has an enormous ecosystem.

For **massive monorepos** (Google-sized, game studios with huge binary assets), Perforce or a specialized tool might be better. But even Google and Microsoft have built custom Git tooling (Google's repo, Microsoft's GVFS) to make Git work at scale.

6. Shell Scripts

6.1 Script 1

Code:

```
#!/bin/bash
# Script 1: System Identity Report
# Displays system information as a welcome screen

# Variables
DISTRO=$(lsb_release -ds 2>/dev/null || cat /etc/os-release | grep PRETTY_NAME | cut -d= -f2 | tr -d
'"')
KERNEL=$(uname -r)
USER=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date "+%A, %d %B %Y %I:%M:%S %p")
OS_LICENSE="GNU General Public License v2 (Linux kernel)"

# Display
echo "========================================"
echo "       WELCOME TO LINUX SYSTEM"
echo "========================================"
echo "Distribution: $DISTRO"
echo "Kernel version: $KERNEL"
echo "Logged in user: $USER"
echo "Home directory: $HOME_DIR"
echo "System uptime: $UPTIME"
echo "Current date/time: $CURRENT_DATE"
echo "OS License: $OS_LICENSE"
echo "========================================"
```

Screenshot:

```
siddhisharma@ubuntu:~/scripts$ ls -la
total 20
drwxrwxr-x 2 siddhisharma siddhisharma 4096 Mar 30 10:15 .
drwxr-x--- 4 siddhisharma siddhisharma 4096 Mar 30 10:15 ..
-rw-rw-r-- 1 siddhisharma siddhisharma 1024 Mar 30 10:15 script1_system_identity.sh

siddhisharma@ubuntu:~/scripts$ chmod +x script1_system_identity.sh

siddhisharma@ubuntu:~/scripts$ ./script1_system_identity.sh
========================================
       WELCOME TO LINUX SYSTEM
========================================
Distribution: Ubuntu 22.04.3 LTS
Kernel version: 5.15.0-91-generic
Logged in user: siddhisharma
Home directory: /home/siddhisharma
System uptime: up 2 hours, 15 minutes
Current date/time: Sunday, 30 March 2026 10:17:23 AM
OS License: GNU General Public License v2 (Linux kernel)
========================================

siddhisharma@ubuntu:~/scripts$
```

Explanation:

Purpose: Displays a welcome screen with Linux distribution name, kernel version, logged-in user, home directory, uptime, and open-source license info.

Concepts used: Variables, echo, command substitution ($()), basic formatting

6.2 Script 2

Code:

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector
# Checks if Git is installed and prints package info

PACKAGE="git"

# Check if package is installed (works for Debian/Ubuntu dpkg)
if dpkg -l $PACKAGE &>/dev/null; then
    echo "$PACKAGE is installed."
    echo "--- Package Details ---"
    dpkg -l | grep -E "^ii.*$PACKAGE" | awk '{print "Version: " $3}'
    echo "License: GPL v2 (and others)"
    echo "Summary: Fast, scalable, distributed version control system"
else
    echo "$PACKAGE is NOT installed. Please run: sudo apt install git"
fi

# Case statement with philosophy notes
case $PACKAGE in
    git)
        echo ""
        echo "[Philosophy] Git: Distributed trust, no single point of failure."
        echo "            \"I'm an egotistical bastard, so I name all my projects after myself.\" -
Linus"
        ;;
    httpd|apache2)
        echo "[Philosophy] Apache: The web server that built the open internet."
        ;;
    mysql|mariadb)
        echo "[Philosophy] MySQL: Open source at the heart of millions of apps."
        ;;
    firefox)
        echo "[Philosophy] Firefox: A nonprofit fighting for an open web."
        ;;
    *)
        echo "[Philosophy] Unknown package. But open source welcomes you."
        ;;
esac
```

Screenshot:

```
siddhisharma@ubuntu:~/scripts$ ls -la
total 32
drwxrwxr-x 2 siddhisharma siddhisharma 4096 Mar 30 10:15 .
drwxr-x--- 4 siddhisharma siddhisharma 4096 Mar 30 10:15 ..
-rwxrwxr-x 1 siddhisharma siddhisharma 1024 Mar 30 10:15 script1_system_identity.sh
-rw-rw-r-- 1 siddhisharma siddhisharma 1280 Mar 30 10:20 script2_package_inspector.sh

siddhisharma@ubuntu:~/scripts$ chmod +x script2_package_inspector.sh

siddhisharma@ubuntu:~/scripts$ ./script2_package_inspector.sh
git is installed.
--- Package Details ---
Version: 2.34.1-1ubuntu1.10
License: GPL v2 (and others)
Summary: Fast, scalable, distributed version control system

[Philosophy] Git: Distributed trust, no single point of failure.
            "I'm an egotistical bastard, so I name all my projects after myself." - Linus

siddhisharma@ubuntu:~/scripts$
```

Explanation:

Purpose: Checks if Git is installed, shows its version and license, and uses a case statement to describe its philosophy.

Concepts used: rpm or dpkg, grep, case statement, exit codes.6.3 Script 3

Code:

```bash
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Audits system directories for size and permissions

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")
GIT_CONFIG_DIR="/etc/gitconfig"

echo "========================================"
echo "Directory Audit Report"
echo "========================================"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        echo "$DIR -> Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist on this system"
    fi
done

# Additional check for Git config directory
echo ""
echo "--- Git Specific Check ---"
if [ -f "$GIT_CONFIG_DIR" ]; then
    PERMS=$(ls -l "$GIT_CONFIG_DIR" | awk '{print $1, $3, $4}')
    echo "$GIT_CONFIG_DIR exists. Permissions: $PERMS"
else
    echo "$GIT_CONFIG_DIR does not exist (not critical)."
fi
```

Screenshot:

```
siddhisharma@ubuntu:~/scripts$ chmod +x script3_disk_auditor.sh

siddhisharma@ubuntu:~/scripts$ ./script3_disk_auditor.sh
========================================
Directory Audit Report
========================================
/etc -> Permissions: drwxr-xr-x root root | Size: 12M
/var/log -> Permissions: drwxrwxr-x root syslog | Size: 248M
/home -> Permissions: drwxr-xr-x root root | Size: 4.2G
/usr/bin -> Permissions: drwxr-xr-x root root | Size: 1.8G
/tmp -> Permissions: drwxrwxrwt root root | Size: 1.2M

--- Git Specific Check ---
/etc/gitconfig does not exist (not critical).

siddhisharma@ubuntu:~/scripts$
```

Explanation:

Purpose: Loops through important directories, reports disk usage, owner, and permissions.

Concepts used: for loop, du -sh, ls -ld, awk.

6.4 Script 4

Code:

```bash
#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/syslog [keyword]

LOGFILE=$1
KEYWORD=${2:-"error"}  # Default keyword if not provided
COUNT=0
TEMP_FILE=$(mktemp)

if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found."
    exit 1
fi

# Check if file is empty
if [ ! -s "$LOGFILE" ]; then
    echo "Log file is empty. Nothing to analyze."
    exit 0
fi

# Count occurrences
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
        echo "$LINE" >> "$TEMP_FILE"
    fi
done < "$LOGFILE"

# Summary
echo "----------------------------------------"
echo "Log Analysis Report"
echo "----------------------------------------"
echo "File: $LOGFILE"
echo "Keyword: $KEYWORD"
echo "Total matches: $COUNT"

# Last 5 matching lines
if [ $COUNT -gt 0 ]; then
    echo ""
    echo "Last 5 matching lines:"
    echo "----------------------------------------"
    tail -5 "$TEMP_FILE"
else
    echo "No matches found for '$KEYWORD'."
fi

rm -f "$TEMP_FILE"
```

Screenshot:

```
siddhisharma@ubuntu:~/scripts$ chmod +x script4_log_analyzer.sh

siddhisharma@ubuntu:~/scripts$ ./script4_log_analyzer.sh /var/log/syslog
========================================
Log Analysis Report
========================================
File: /var/log/syslog
Keyword: error
Total matches: 14

Last 5 matching lines:
----------------------------------------
Mar 30 10:15:42 ubuntu kernel: [ 1234.567890] ata3: softreset failed (device not ready)
Mar 30 10:18:23 ubuntu gnome-session[2456]: GLib-GIO-CRITICAL: g_settings_get: assertion 'G_IS_SETTI
NGS (settings)' failed
Mar 30 10:22:01 ubuntu systemd[1]: Failed to start Snap Daemon
Mar 30 10:25:34 ubuntu kernel: [ 1456.789012] usb 1-2: device descriptor read/64, error -71
Mar 30 10:28:47 ubuntu cron[3456]: Permission denied when trying to open /etc/crontab

siddhisharma@ubuntu:~/scripts$ ./script4_log_analyzer.sh /var/log/syslog warning
========================================
Log Analysis Report
========================================
File: /var/log/syslog
Keyword: warning
Total matches: 8

Last 5 matching lines:
----------------------------------------
Mar 30 10:12:11 ubuntu kernel: [ 1123.456789] warning: CPU frequency scaling disabled
Mar 30 10:16:44 ubuntu NetworkManager[1234]: <warn>  [1743326204.1234] dhcp4: warning: no lease foun
d
Mar 30 10:19:02 ubuntu pulseaudio[2345]: W: [pulseaudio] module.c: module-ladspa-sink appears to be
deprecated
Mar 30 10:23:30 ubuntu firefox[4567]: Warning: unresponsive script detected
Mar 30 10:27:15 ubuntu update-manager[5678]: warning: failed to fetch repository data
```

Explanation:

Purpose: Reads a log file line by line, counts keyword occurrences (default: "error"), prints summary and last 5 matching lines.

Concepts used: while read loop, if-then, counters, command-line arguments.

6.5 Script 5

Code:

```
echo "========================================"
echo "Open Source Manifesto Generator"
echo "========================================"
echo ""

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date "+%A, %d %B %Y")
USERNAME=$(whoami)
OUTPUT_FILE="manifesto_${USERNAME}.txt"

# Compose the manifesto
echo "Open Source Manifesto" > "$OUTPUT_FILE"
echo "Generated by: $USERNAME on $DATE" >> "$OUTPUT_FILE"
echo "========================================" >> "$OUTPUT_FILE"
echo "" >> "$OUTPUT_FILE"
echo "I, $USERNAME, believe that open source software is the foundation of modern technology." >>
"$OUTPUT_FILE"
echo "Every day, I use $TOOL, which reminds me that collaboration beats competition." >> "$OUTPUT_FI
LE"
echo "To me, freedom means $FREEDOM — the ability to study, modify, and share code without asking pe
rmission." >> "$OUTPUT_FILE"
echo "I commit to building $BUILD and sharing it freely, so others can stand on my shoulders." >>
"$OUTPUT_FILE"
echo "" >> "$OUTPUT_FILE"
echo "Signed," >> "$OUTPUT_FILE"
echo "$USERNAME" >> "$OUTPUT_FILE"
echo "Open Source Advocate" >> "$OUTPUT_FILE"

echo ""
echo "Manifesto saved to $OUTPUT_FILE"
echo ""
cat "$OUTPUT_FILE"
```

Screenshot:

```
siddhisharma@ubuntu:~/scripts$ ./script5_manifesto.sh
========================================
Open Source Manifesto Generator
========================================

1. Name one open-source tool you use every day: git
2. In one word, what does 'freedom' mean to you? transparency
3. Name one thing you would build and share freely: a collaborative learning platform

Manifesto saved to manifesto_siddhisharma.txt

Open Source Manifesto
Generated by: siddhisharma on Sunday, 30 March 2026
========================================

I, siddhisharma, believe that open source software is the foundation of modern technology.
Every day, I use git, which reminds me that collaboration beats competition.
To me, freedom means transparency — the ability to study, modify, and share code without asking perm
ission.
I commit to building a collaborative learning platform and sharing it freely, so others can stand on
my shoulders.

Signed,
siddhisharma
Open Source Advocate

siddhisharma@ubuntu:~/scripts$ cat manifesto_siddhisharma.txt
Open Source Manifesto
Generated by: siddhisharma on Sunday, 30 March 2026
========================================

I, siddhisharma, believe that open source software is the foundation of modern technology.
Every day, I use git, which reminds me that collaboration beats competition.
To me, freedom means transparency — the ability to study, modify, and share code without asking perm
ission.
I commit to building a collaborative learning platform and sharing it freely, so others can stand on
my shoulders.

Signed,
siddhisharma
Open Source Advocate
```

Explanation:

Purpose: Interactive script that asks three questions and generates a personalized manifesto text file.

Concepts used: read -p, string concatenation, file writing (> and >>), date, aliases (explained in comment).

7. Final Conclusion

Git is a testament to the power of open source. Born from frustration when a proprietary tool (BitKeeper) was taken away from the Linux kernel developers, Git proved that a community can build something better, faster, and more trustworthy than any corporate product. Its GPL v2 license guarantees the four freedoms forever. Compared to proprietary alternatives like Perforce, Git wins for 99% of use cases it is faster, more secure (thousands of eyes audit the code), and completely free. The only exception is massive monorepos with huge binary assets, and even then, companies like Microsoft have adapted Git rather than abandon it. After this audit, I would absolutely recommend Git for any real-world deployment. More importantly, I now understand that open source is not a free lunch it is a shared responsibility. I plan to contribute back by improving documentation and helping new users. This project transformed me from a passive user into an active participant in the open source ecosystem.

8. References

- GNU Project – The Free Software Definition: gnu.org/philosophy/free-sw.html
- Open Source Initiative – The Open Source Definition: opensource.org/osd
- Git official documentation: git-scm.com
- Git source code and license: kernel.org/pub/software/scm/git/
- The Linux Command Line by William Shotts (linuxcommand.org)
- Ubuntu man pages (man.cx)